COMP 4026 proposal

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1. Introduction and Project Objective

Over the past decade, there has been remarkable progress in facial analysis technologies [1], among which facial expression recognition [2] and face recognition [3] are two tasks with the most advanced studies and commercial deployment. However, the widespread deployment of such systems raises significant privacy concerns, as facial images encode sensitive personal information. As a result face anonymisation has emerged as one of the solutions [4]. The core idea is to transform a facial image so that the individual's identity can not be recognised, while retaining other useful attributes like the facial expression.

This project aims to build an integrated system for anonymised facial expression recognition. The system consists of three modules: Face Recognition Module, Face Anonymisation Module and Facial Expression Recognition Module. The goal is that the anonymised face should not be identified by the recogniser while preserving expression classification accuracy. The system will be evaluated using face recognition accuracy as the privacy metric and expression consistency rate as the utility metric.

2.Preliminary Studies

In the preliminary study, we reviewed representative methods for the three main parts of the project: face recognition, face anonymisation, and facial expression recognition. For face recognition, Viola-Jones and Eigenfaces are suitable baseline methods because they are classical, efficient, and easy to integrate into a prototype system [5], [6]. For face anonymisation, simple methods such as Gaussian blur and pixelation are easy to implement, but they may remove useful facial details together with identity information. Therefore, a generative method is more suitable for this project, since it aims to hide identity while preserving the overall facial structure [7]. For facial expression recognition, FER-2013 is a widely used benchmark dataset and provides a practical basis for training and evaluation [8]. Based on this preliminary study, our project uses simple baseline methods where appropriate and adopts a more suitable anonymisation direction to better balance privacy protection and expression preservation.

3.Proposed Methodology

3.1 Face recognition

We plan to make facial recognition a fundamental part of identity analysis, maintaining high accuracy before anonymization and lower accuracy after anonymization.

Firstly, we use Viola-Jones face detector to extract face ROIs from the input image to reduce background interference. Then, trainingEigenface-based recognizer on the Pins Face Recognition dataset to use the processed face images, and test on both original and anonymized face images to support subsequent privacy-related evaluation.

3.2 Face anonymisation

For the face anonymisation module, we plan to use Gaussian blur and pixelation as simple baselines, and explore a GAN-based anonymisation approach as the main direction. This is because our project requires not only privacy protection, but also the preservation of facial expressions after anonymisation. This module will be trained and tested on CelebA-HQ, which is the dataset specified for the face anonymisation task in the project brief and already adopted in our initial design. Its effectiveness will later be evaluated in the integrated system by checking whether face recognition accuracy drops on anonymised images while expression consistency is still maintained.

3.3 Facial expression recognition

The facial expression recognition module serves as the utility evaluator of the system. Its primary task is to classify facial images into one of seven expressions defined by the FER-2013 dataset. By comparing the expression labels of the original face images against those predicted on anonymised versions, we can quantify whether the anonymisation successfully preserves facial expression information. The key output metric is the Expression Consistency Rate, defined as the proportion for which the predicted expression remains unchanged after anonymisation.

We plan to adopt a transfer learning approach based on convolutional neural networks like ResNet-18 as our baseline due to its effectiveness and wide adoption in image classification tasks. If time and resources permit, we will also explore more powerful methods such as ResNet-50 or EfficientNet-B0 and compare their performance to identify the most suitable architecture.

4.Initial System Design

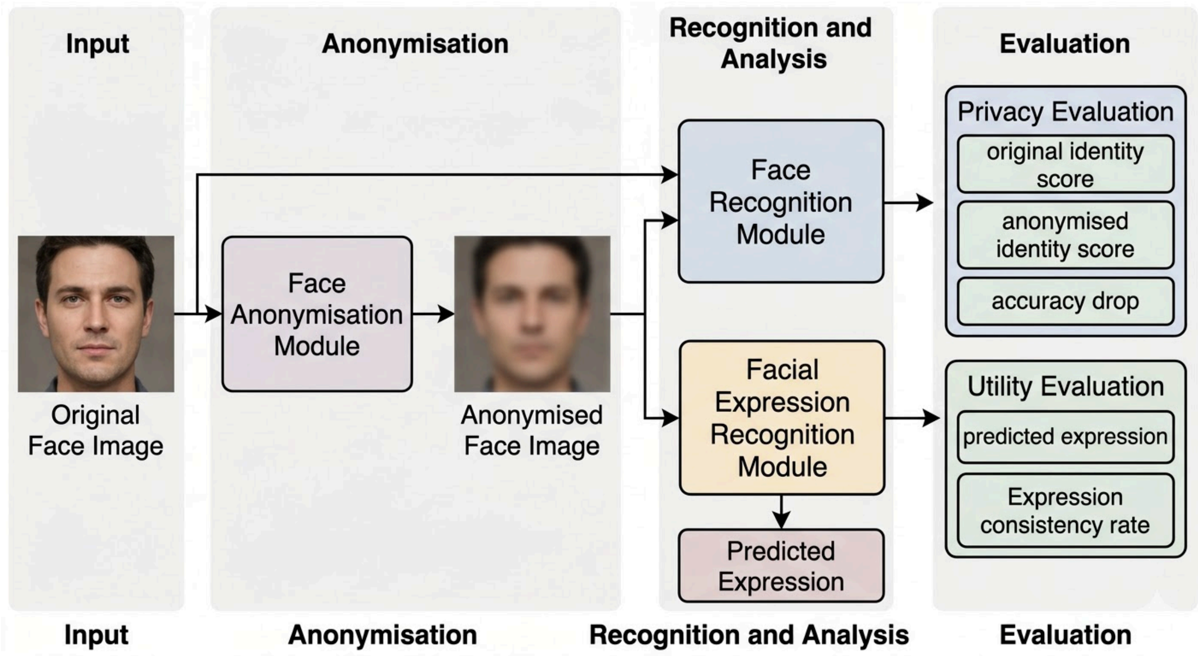

Figure 1. Initial System Design of the Proposed Anonymised Facial Expression Recognition System

Figure 1 demonstrates the overall pipeline of the proposed system. The system starts from an original face image, performing face anonymisation to generate an anonymised version. Then, it passes to the recognition and analysis phase shown in the figure. This phase has two branches: one focuses on identity-related analysis, and the other focuses on facial expression analysis. Ultimately, each branch confirms the system's effectiveness through their own evaluations. This design reflects the main goal of the project, which is to reduce identity recognisability while preserving facial expression information.

5.Datasets and Evaluation Metrics

5.1Datasets for Training:

We use the dataset required by the course materials.

Face Anonymization Model: CelebA-HQ. A clean, high-quality datasets are suitable for face image transformation.

Expression Classification: FER-2013. The dataset with emoji tags is suitable for emoji recognition.

Face Recognition Dataset: Pins Face Recognition. The dataset categorized by person and their identity is suitable for person recognition.

5.2 Evaluation Metrics

5.2.1 Module-level evaluation

This section verifies the training effectiveness of each module during the training module. For face recognition, we will examine the identity recognition performance on the original face dataset. For expression detection, we will examine the classification performance on the expression dataset.

5.2.2 Integrated system evaluation

After module development, the three modules will be integrated into a unified pipeline.

We select a set of face images for final evaluation, then anonymize these images and get anonymised versions.

Privacy will be measured by the drop in face recognition accuracy, while utility will be measured by expression consistency.

6. Work Allocation

Member	Role	Primary Responsibilities
LU Xuanchen	Face recognition	Develop and train a face recognition model on the Pins Face Recognition dataset to serve as the privacy evaluator.
JIA Zaixi	Face anonymisation	Develop a face anonymisation model on CelebA-HQ to generate de-identified images that preserve facial expressions.
LIU Jicang	Expression recognition	Train a facial expression classifier on FER-2013 to serve as the utility evaluator.
All	Integration & Evaluation	Integrate all modules into an end-to-end pipeline, conduct joint experiments, and prepare the presentation and report.

References

[1] A. Kortylewski, B. Egger, A. Schneider, T. Gerig, A. Morel-Forster, and T. Vetter, "Analyzing and reducing the damage of dataset bias to face recognition with synthetic data," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2019, pp. 2261–2268.

[2] S. Li and W. Deng, "Deep facial expression recognition: A survey," IEEE Trans. Affective Comput., vol. 13, no. 3, pp. 1195–1215, 2022.

[3] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 4690–4699.

[4] T. Li and L. Lin, "AnonymousNet: Natural face de-identification with measurable privacy," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2019, pp. 56–65.

[5] P. Viola and M. Jones, “Rapid object detection using a boosted cascade of simple features,” in *Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, Kauai, HI, USA, 2001, vol. 1, pp. I-511–I-518.

[6] M. Turk and A. Pentland, “Eigenfaces for recognition,” *Journal of Cognitive Neuroscience*, vol. 3, no. 1, pp. 71–86, 1991.

[7] H. Hukkelås, R. Mester, and F. Lindseth, “DeepPrivacy: A generative adversarial network for face anonymization,” in *Advances in Visual Computing*, 2019, pp. 565–578.

[8] I. J. Goodfellow *et al.*, “Challenges in representation learning: A report on three machine learning contests,” *Neural Networks*, vol. 64, pp. 59–63, 2015

[9] P. Viola and M. Jones, “Robust Real-Time Face Detection,” *International Journal of Computer Vision*, 2004